

Split AC Systems

Components | Refrigerant Circuit | Inverter Tech | Selection

What You Will Learn

- How a split AC system is built: indoor unit, outdoor unit, refrigerant circuit, drain, controls
- The five common types: wall-mounted, floor-standing, cassette, ducted, multi-split
- Inverter vs fixed-speed compressors — how each affects comfort, energy use and lifespan
- Performance ratings (EER, COP, SEER, ISEER) and how to size a unit correctly

Energy Efficiency Ratio (cooling performance)

$$\text{EER} = \text{Cooling Output (W)} \div \text{Power Input (W)}$$

Higher EER/COP = more cooling per watt of electricity = lower running cost

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HVAC-011: Split AC Systems

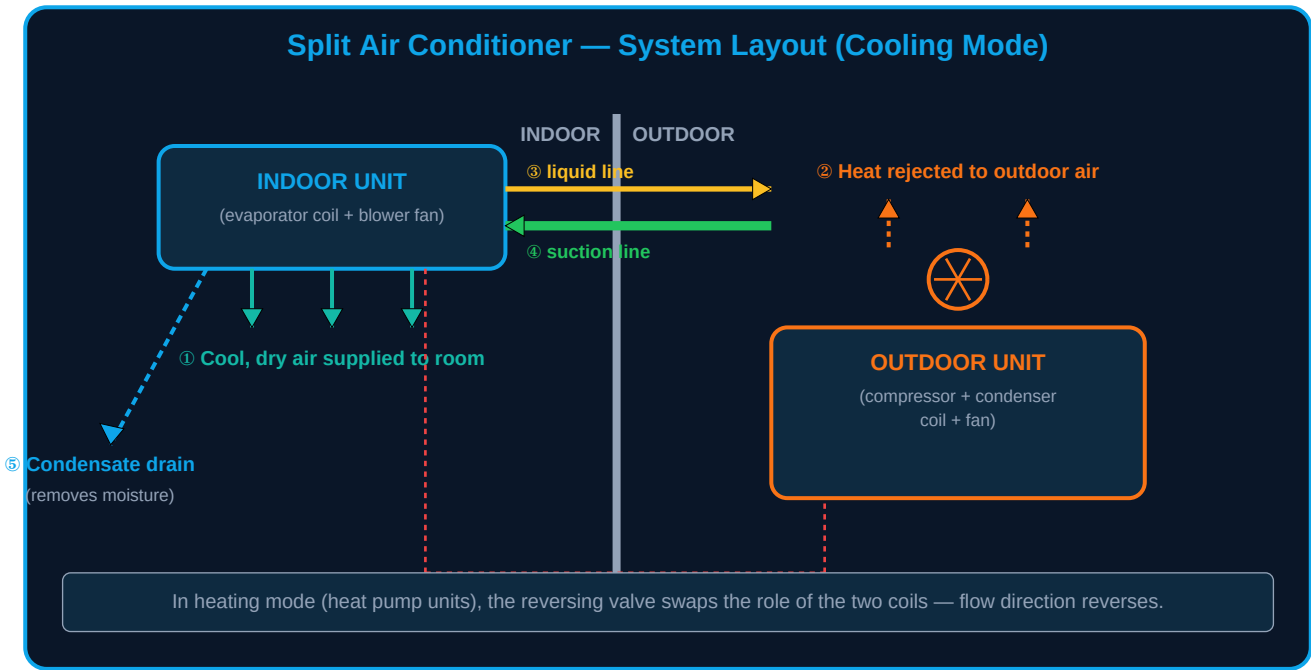
A **split air conditioner** is the most common cooling system found in homes, offices, shops, and small commercial spaces worldwide. It gets its name from the fact that the system is "split" into two parts — an indoor unit that cools the room, and an outdoor unit that rejects the absorbed heat to the outside air — connected by a sealed refrigerant circuit. Understanding how these two halves work together is the foundation for everything else in residential and light commercial HVAC: selection, installation, and troubleshooting.

Learning Objectives

- Identify the major components of a split AC system and explain what each one does
- Trace the refrigerant flow between the indoor and outdoor units in cooling mode
- Compare the five common split system types and where each is best applied
- Explain the difference between inverter and fixed-speed compressors and why it matters
- Read EER, COP, SEER, and ISEER ratings and use them to compare unit efficiency
- Apply basic sizing guidelines to select an appropriately-rated unit for a room

1. Anatomy of a Split AC System

Every split system — no matter how large or small — is built from the same six functional parts. The diagram below shows how they connect during normal cooling operation.



#	Component	Function
①	Indoor unit (evaporator)	Contains the evaporator coil and a blower fan. Warm room air is drawn across the cold coil — heat and moisture are absorbed into the refrigerant, and cool, dehumidified air is blown back into the room.

②	Outdoor unit (condenser)	Houses the compressor, the condenser coil, and a fan. The compressor raises the refrigerant pressure and temperature; the fan blows outdoor air across the condenser coil to reject the absorbed heat.
③	Liquid line	Smaller-diameter copper pipe carrying high-pressure liquid refrigerant from the outdoor unit to the indoor unit's expansion device.
④	Suction line	Larger-diameter, insulated copper pipe returning low-pressure refrigerant gas from the indoor evaporator back to the compressor.
⑤	Condensate drain	Plastic pipe that carries away the water condensed out of the room air on the cold evaporator coil — must always run with a continuous downward slope.
⑥	Power & control cable	Supplies electrical power to the outdoor unit and carries control signals between indoor and outdoor units (and to the remote/thermostat).

2. The Refrigerant Circuit in Cooling Mode

The split system works by continuously moving a refrigerant — a fluid that changes between liquid and gas at convenient temperatures — around a sealed loop. In the indoor unit, the refrigerant absorbs heat from room air and evaporates into a low-pressure gas. The compressor in the outdoor unit then squeezes that gas to a high pressure and temperature. As outdoor air blows across the condenser coil, the hot gas gives up its heat and condenses back into a high-pressure liquid. An expansion device drops the pressure sharply just before the liquid re-enters the indoor coil, where the cycle repeats. This is the same basic refrigeration cycle covered in HVAC-002 — a split AC is simply that cycle packaged into two connected boxes.

Operating Modes Found on Most Split AC Remote Controls

Mode	What Happens	Typical Use
Cool	Refrigerant cycle runs as described above — evaporator absorbs heat indoors, condenser rejects it outdoors.	Standard daytime/summer operation
Heat (heat pump units only)	A reversing valve swaps the role of the two coils — the indoor coil becomes the condenser (gives off heat) and the outdoor coil becomes the evaporator (absorbs heat from outside air).	Cold-season heating without a separate furnace
Dry / Dehumidify	Compressor and fan run at reduced speed so the coil stays cold enough to remove moisture without over-cooling the space.	Humid weather, minimal temperature drop wanted
Fan only	Indoor blower runs continuously; compressor stays off — air circulates but is not cooled.	Air movement without running the compressor
Auto	The unit automatically switches between cooling and (on heat pump models) heating to hold the set temperature.	Hands-off comfort control

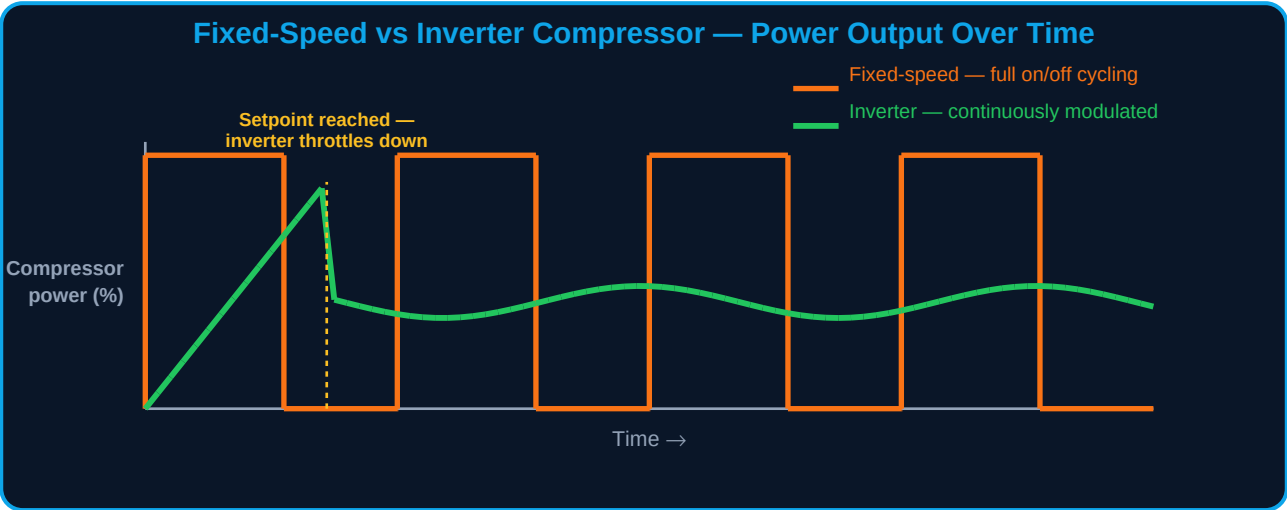
3. Five Common Types of Split AC Systems

"Split AC" is a family of system layouts, not a single product. The right type for a project depends on the building layout, ceiling space, budget, and how many rooms need conditioning from one outdoor unit.

Type	Description	Best Applications
Wall-mounted	The classic compact indoor unit fixed high on an interior wall. Easiest and cheapest to install.	Bedrooms, single offices, shops — single-room cooling
Floor-standing / console	Taller cabinet unit installed at floor or low-wall level, discharging air upward and outward.	Rooms without suitable high wall space; retrofits
Ceiling cassette	Square unit recessed into a suspended ceiling, distributing air in four directions for even coverage.	Open-plan offices, retail spaces, restaurants
Ducted (concealed)	Indoor unit hidden above the ceiling, connected to a network of ducts and grilles — no visible indoor unit.	Whole-apartment or whole-floor cooling with a clean appearance
Multi-split	One outdoor unit supplies refrigerant to two or more indoor units (any of the types above), each controlled independently.	Apartments and small offices needing several rooms cooled from one outdoor unit

4. Inverter vs Fixed-Speed Compressors

The compressor is the "heart" of the system, and how it is controlled has a bigger impact on comfort, running cost, and equipment lifespan than almost any other design choice. There are two fundamentally different approaches on the market today.



A **fixed-speed** (also called "on/off" or "non-inverter") compressor can only run at one speed — full power — or be switched off completely. To hold a room at the set temperature, it must repeatedly switch on, overshoot the setpoint, switch off, let the temperature drift back up, then switch on again. Each restart draws a large inrush current and stresses the motor and refrigerant circuit.

An **inverter** compressor uses a variable-frequency drive to continuously adjust its motor speed — running fast when a large amount of cooling is needed (e.g. when first switched on) and then throttling down to a low, steady speed once the room reaches the setpoint, matching output to the actual load moment by moment. This avoids hard cycling entirely.

Aspect	Fixed-Speed Compressor	Inverter Compressor
Temperature stability	Swings 2–3°C above/below setpoint as it cycles on and off	Holds within ≈0.5°C of setpoint once stabilised
Energy consumption	Higher — full power even when only partial cooling is needed, plus restart losses	Lower — typically 30–50% less for the same comfort level
Starting current / inrush	High inrush every time it restarts — stresses wiring and the motor	Soft, gradual ramp-up — far gentler on components and the electrical supply
Noise	Noticeable jump in noise each time the compressor kicks in	Smooth, generally quieter operation at part-load
Purchase price	Lower upfront cost	Higher upfront cost — typically recovered through energy savings over a few years
Typical lifespan	Shorter — frequent on/off cycling accelerates wear	Longer — running at lower, steadier speeds reduces mechanical stress

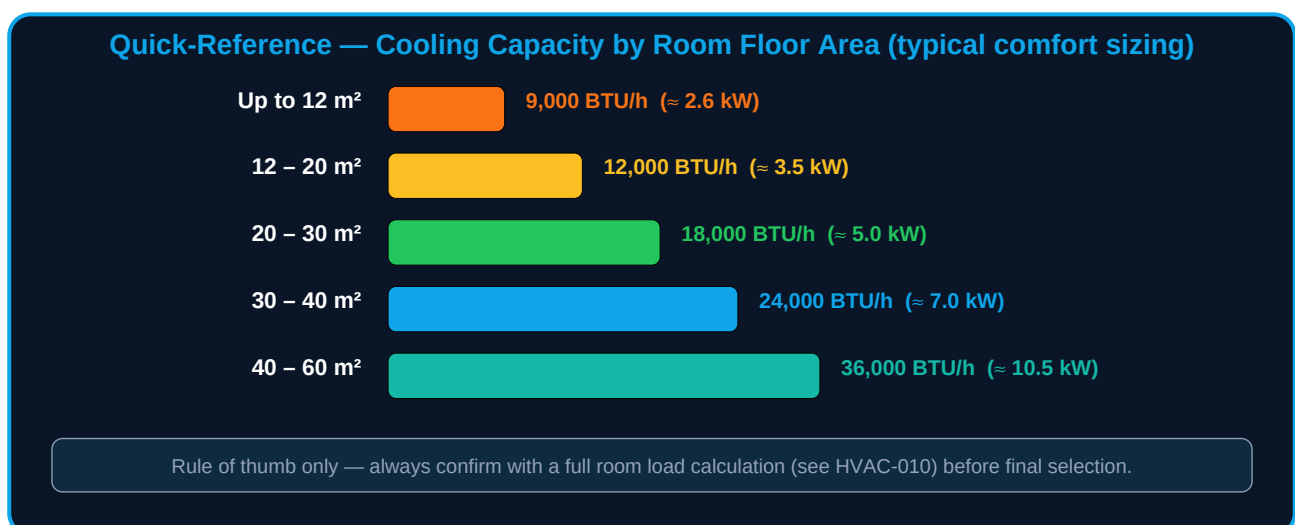
5. Performance Ratings — How to Compare Units

Manufacturers publish standardised efficiency ratings so that buyers can compare how much cooling (or heating) a unit delivers per unit of electricity it consumes. A higher rating always means lower running costs for the same comfort level — this is often more important over the equipment's lifetime than the purchase price.

Rating	Stands For	What It Tells You
EER	Energy Efficiency Ratio	Cooling output (W) ÷ electrical power input (W) at one fixed test condition. Simple instantaneous efficiency snapshot.
COP	Coefficient of Performance	Same idea as EER but expressed as a dimensionless ratio (output ÷ input in matching units) — commonly used for both cooling and heating performance.
SEER	Seasonal Energy Efficiency Ratio	Average efficiency over a full cooling season, accounting for the unit running at partial load most of the time. A far more realistic real-world figure than EER.
ISEER	Indian Seasonal Energy Efficiency Ratio	A regional seasonal rating system (used on the Indian subcontinent and increasingly referenced elsewhere) similar in purpose to SEER.

6. Sizing and Selection Guidelines

Selecting a split AC unit always starts with a proper room cooling load calculation (see HVAC-010) — guessing leads to the same problems described there: discomfort, wasted energy, or poor humidity control. The chart below gives a rough starting point for quick estimates and sanity-checking a calculated result.



Beyond raw capacity, several practical placement and installation factors strongly affect how well a split system performs once it is running:

- **Indoor unit height:** mount wall units roughly 2.1–2.4 m above floor level for even air distribution and to keep the discharge airflow off occupants directly below.

- **Outdoor unit clearance:** leave at least 30–60 cm of open space around the condenser coil and fan discharge — a blocked outdoor unit cannot reject heat properly and the system will struggle to cool.
- **Pipe run length:** every split system has a maximum rated refrigerant pipe length and a maximum height difference between indoor and outdoor units — exceeding either reduces capacity and can damage the compressor.
- **Drain slope:** the condensate drain line must fall continuously toward its discharge point — even a short flat or upward section will cause water to back up and overflow into the room.

7. Practice Exercises

#	Exercise	Task
1	Component identification	Sketch a split AC layout from memory and label all six components from Section 1. For each one, write one sentence describing its function in your own words.
2	Mode selection	A client complains their room feels clammy but not too hot during the rainy season. Which operating mode from the table in Section 2 would you recommend, and why?
3	Type selection	An open-plan retail shop (10 m × 8 m, suspended ceiling, no spare wall space) needs even cooling without a visible bulky indoor unit. Pick the most suitable type from Section 3 and justify your choice against at least one alternative.
4	Inverter vs fixed-speed comparison	A homeowner is choosing between two units of equal capacity: Unit A (fixed-speed, lower price) and Unit B (inverter, 30% higher price, SEER rating 40% better). Estimate at what point Unit B "pays for itself" if the home runs the AC roughly 6 hours/day for 200 days/year — state your assumptions about electricity cost.
5	Sizing sanity check	A bedroom measures 4.5 m × 5 m. Using the chart in Section 6, what capacity would you expect to need? Now compare that figure with a full HVAC-010 style load calculation for the same room — explain any difference you would expect to see.

Lesson Summary

Key Concept	Summary
Six core components	Indoor unit (evaporator), outdoor unit (compressor + condenser), liquid line, suction line, condensate drain, power/control cable.
Refrigerant circuit	Same basic refrigeration cycle as HVAC-002 — evaporator absorbs heat indoors, compressor raises pressure/temperature, condenser rejects heat outdoors, expansion device drops pressure to repeat the cycle.
Five system types	Wall-mounted, floor-standing/console, ceiling cassette, ducted (concealed), and multi-split — chosen based on layout, ceiling space, budget, and number of rooms.

Inverter vs fixed-speed	Fixed-speed cycles fully on/off and overshoots the setpoint; inverter modulates compressor speed continuously for steadier comfort, lower energy use, and longer equipment life.
Efficiency ratings	EER and COP are instantaneous efficiency snapshots; SEER and ISEER are seasonal averages that better reflect real-world running cost — always compare seasonal ratings when possible.
Sizing approach	Start from a proper room load calculation (HVAC-010); use capacity charts only as a rough sanity check, never as the final basis for selection.
Installation factors	Indoor unit mounting height, outdoor unit airflow clearance, maximum refrigerant pipe length/height difference, and a continuously-sloped condensate drain all affect real-world performance.

What Is Next — HVAC-012

HVAC-012: VRF Systems — How Variable Refrigerant Flow technology scales the split-system concept up to whole buildings: one outdoor condensing unit serving dozens of indoor units, each independently modulated. Heat recovery between zones, piping network design basics, and where VRF fits between split systems and central chilled-water plants.

Resource	Details
HVAC PDF Series	HVAC-001 through HVAC-030 — ayetechub.com/pdfs.html
HVAC Calculator	Free Solar PV & HVAC sizing tools — ayetechub.com/downloads
Community	Telegram: t.me/ayetechub

Quick Glossary — Terms Used in This Lesson

Term	Meaning
Refrigerant	The fluid that circulates through the sealed circuit, absorbing heat indoors as it evaporates and releasing it outdoors as it condenses.
Compressor	The "pump" of the system — raises refrigerant pressure and temperature so it can give up its heat outdoors.
Expansion device	A restriction that drops refrigerant pressure sharply just before it re-enters the indoor coil, allowing it to absorb heat again.
Reversing valve	Found only on heat pump models — swaps the role of the indoor and outdoor coils to switch between cooling and heating.
Inrush current	The brief surge of current a motor draws at the moment it starts — far higher than its normal running current.